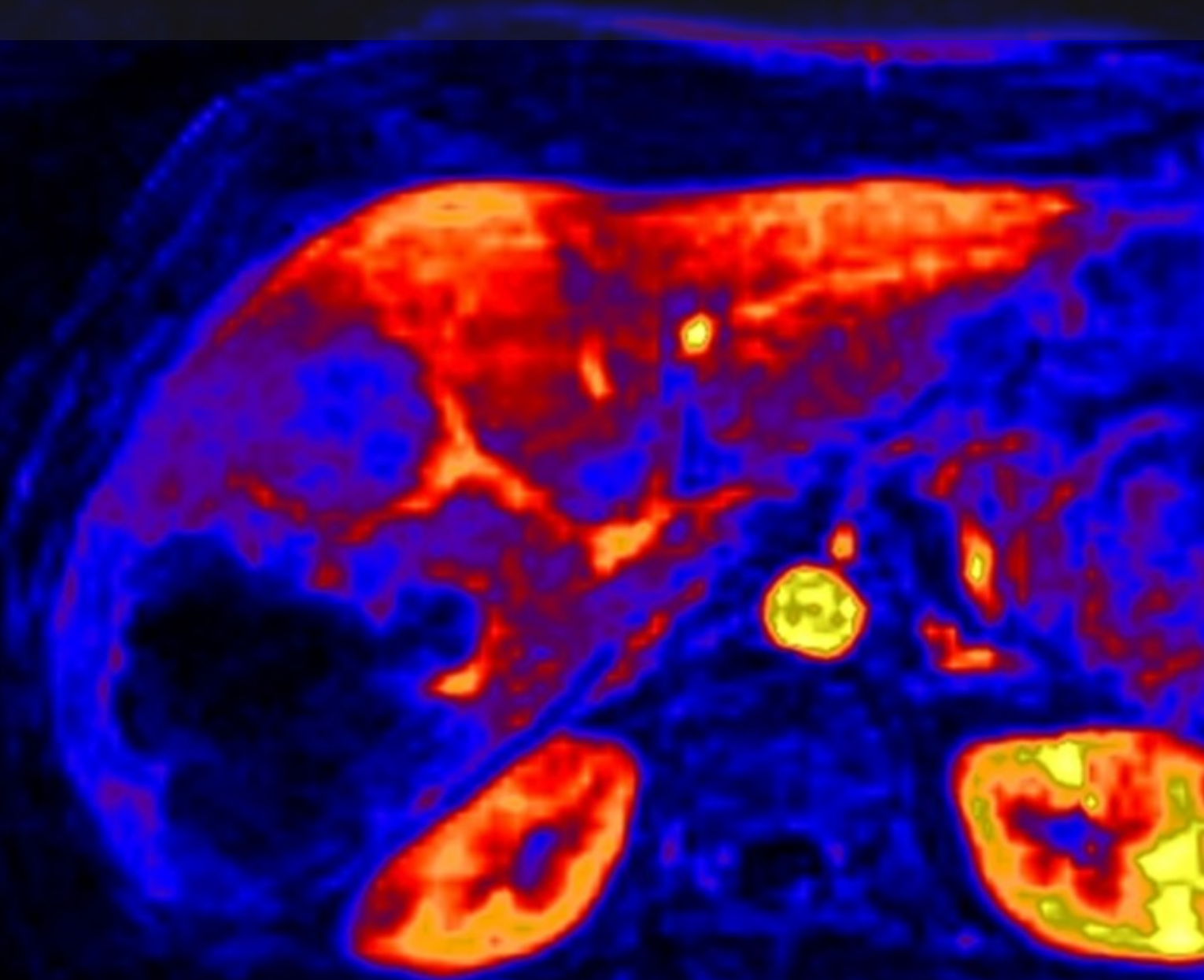


Sheffield two-compound study

metformin (all results)

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metformin (all results)

by

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Report compiled by:	Steven Sourbron
Institute:	University of Sheffield
Department:	Section of Medical Imaging and Technologies
Email:	s.sourbron@sheffield.ac.uk
Date:	Friday 7 th November, 2025



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Two-scan results

1.1. Data summary

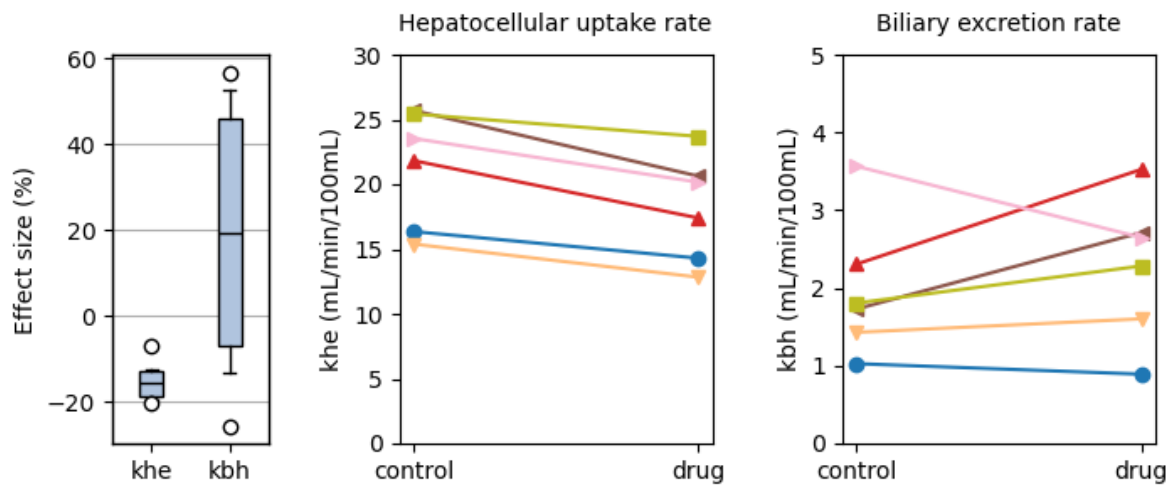


Figure 1.1: Effect size (%) on hepatocellular uptake (k_{he} , left) and biliary excretion (k_{bh} , right) of gadoxetate. The boxplot shows median, interquartile range and 95 percent range. The line plots show individual values for hepatocellular uptake (k_{he} , middle) and biliary excretion (k_{bh} , right) of gadoxetate of the control (left of plot) and treatment (right of plot). Grey lines are healthy controls with rifampicin injection.

parameter	count	mean	std	min	25%	50%	75%	max
khe effect size (%)	6.0	-15.0	5.0	-20.3	-18.9	-15.5	-13.0	-6.8
khe control (mL/min/100cm ³)	6.0	21.37	4.49	15.38	17.72	22.68	24.95	25.69
khe drug (mL/min/100cm ³)	6.0	18.17	4.11	12.83	15.07	18.78	20.52	23.69
kbh effect size (%)	6.0	18.2	33.8	-25.6	-6.8	19.2	46.2	56.8
kbh control (mL/min/100cm ³)	6.0	1.98	0.89	1.03	1.5	1.77	2.18	3.56
kbh drug (mL/min/100cm ³)	6.0	2.28	0.92	0.89	1.77	2.47	2.7	3.52

Table 1.1: Effect size and absolute values of hepatocellular uptake (k_{he}) and biliary excretion (k_{bh}) of gadoxetate

1.2. Liver biomarkers

Biomarker	p-value	Bayes Factor	Odds Ratio
AUC for CI (0-35min)	0.98892	0.37	0.99
AUC for CI (0-inf)	0.52995	0.45	1.94
Biliary excretion rate	0.38756	0.53	0.55
Biliary tissue excretion rate	0.45121	0.48	1.51
Final biliary excretion rate	0.33227	0.58	0.42
Final hepatocellular uptake rate	0.16666	0.89	2.72
Gut dispersion	0.57444	0.43	1.85
Gut mean transit time	0.77053	0.39	1.47
Hepatocellular mean transit time	0.50276	0.46	0.74
Hepatocellular tissue uptake rate	0.17954	0.85	0.35
Hepatocellular uptake rate	0.00202	23.32	3.84
Initial biliary excretion rate	0.84845	0.38	0.88
Initial hepatocellular uptake rate	0.54238	0.44	1.94
Liver T1-MOLLI at 45min	0.49899	0.46	0.65
Liver T1-MOLLI at baseline	0.46924	0.48	1.92
Liver T1-MOLLI at scan 2	0.37867	0.53	3.57
Liver extracellular volume fraction	0.08747	1.4	8.1
Liver plasma clearance	0.00153	28.77	2.5
RE for R11 at 20min	0.8491	0.38	0.85
RE for SI at 20min	0.67919	0.4	0.68

Table 1.2: Results of a pairwise comparison testing for differences in liver biomarkers between control and treatment. The results are ranked by their p-value, with most significant differences at the top of the list.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	202.0 (50.0)	202.0 (23.0)	11.5 (36.0)
AUC for CI (0-inf)	mM*sec	783.0 (360.0)	664.0 (77.0)	38.5 (95.0)
Biliary excretion rate	mL/min/100cm3	1.98 (0.71)	2.28 (0.74)	18.2 (27.0)
Biliary tissue excretion rate	/min	0.0351 (0.011)	0.0325 (0.0082)	-4.68 (16.0)
Final biliary excretion rate	mL/min/100cm3	1.75 (0.6)	2.17 (0.79)	28.0 (43.0)
Final hepatocellular uptake rate	mL/min/100cm3	25.5 (8.9)	20.7 (4.2)	-11.9 (18.0)
Gut dispersion	%	72.2 (5.2)	69.4 (7.8)	-3.27 (12.0)
Gut mean transit time	sec	21.5 (2.5)	20.7 (3.4)	-1.28 (21.0)
Hepatocellular mean transit time	min	32.0 (9.4)	33.9 (9.6)	8.97 (19.0)
Hepatocellular tissue uptake rate	/min	0.935 (0.29)	1.19 (0.42)	32.1 (43.0)
Hepatocellular uptake rate	mL/min/100cm3	21.4 (3.6)	18.2 (3.3)	-15.0 (4.0)
Initial biliary excretion rate	mL/min/100cm3	2.36 (0.98)	2.43 (0.75)	11.1 (31.0)
Initial hepatocellular uptake rate	mL/min/100cm3	17.3 (4.2)	15.7 (2.5)	2.1 (40.0)
Liver T1-MOLLI at 45min	sec	0.445 (0.062)	0.47 (0.098)	5.58 (13.0)
Liver T1-MOLLI at baseline	sec	0.761 (0.025)	0.75 (0.023)	-1.31 (3.5)
Liver T1-MOLLI at scan 2	sec	0.621 (0.076)	0.568 (0.041)	-5.77 (18.0)
Liver extracellular volume fraction	mL/100cm3	24.3 (3.9)	17.4 (5.6)	-26.7 (23.0)
Liver plasma clearance	L/min	0.212 (0.054)	0.177 (0.056)	-17.9 (5.9)
RE for R11 at 20min	%	71.4 (23.0)	73.3 (11.0)	27.7 (60.0)
RE for SI at 20min	%	61.7 (20.0)	65.7 (7.4)	42.2 (86.0)

Table 1.3: Mean values along with their 95 percent confidence intervals for all liver biomarkers of the control and treatment visit. The last column shows the relative change at the treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

1.3. Systemic biomarkers

Biomarker	p-value	Bayes Factor	Odds Ratio
AUC for Cb (0-35min)	0.29871	0.61	0.39
AUC for Cb (0-inf)	0.82527	0.38	1.28
Body extraction fraction	0.55466	0.44	0.48
Cardiac output	0.14992	0.96	2.41
Heart-lung dispersion	0.50784	0.46	0.63
Heart-lung mean transit time	0.55015	0.44	0.44
Organs blood mean transit time	0.73826	0.39	0.71
Organs extraction fraction	0.33669	0.57	0.24
Organs extravascular mean transit time	0.14619	0.98	4.76
RE for R1b at 20min	0.54583	0.44	0.57
RE for Sb at 20min	0.30243	0.61	2.3

Table 1.4: Results of a pairwise comparison testing for differences in systemic biomarkers between control and treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	35.7 (9.1)	41.7 (9.1)	27.2 (41.0)
AUC for Cb (0-inf)	mM*sec	76.3 (36.0)	71.6 (14.0)	36.8 (85.0)
Body extraction fraction	%	4.62 (2.1)	5.45 (1.1)	76.9 (110.0)
Cardiac output	L/min	8.15 (2.9)	6.59 (2.2)	-17.2 (21.0)
Heart-lung dispersion	%	43.9 (6.8)	45.9 (5.6)	6.56 (15.0)
Heart-lung mean transit time	sec	18.4 (3.5)	20.0 (2.3)	16.2 (32.0)
Organs blood mean transit time	sec	25.4 (4.9)	27.0 (8.5)	12.1 (41.0)
Organs extraction fraction	%	18.0 (2.5)	21.1 (3.7)	23.3 (39.0)
Organs extravascular mean transit time	min	8.74 (2.1)	6.91 (1.2)	-16.4 (22.0)
RE for R1b at 20min	%	14.1 (4.5)	15.6 (3.5)	28.9 (53.0)
RE for Sb at 20min	%	16.4 (7.3)	13.2 (2.8)	-4.22 (30.0)

Table 1.5: Mean values along with their 95 percent confidence intervals for all systemic biomarkers at the control and treatment visit. The last column shows the relative change at the treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

1.4. Case notes
1.4.1. Subject SHF-003

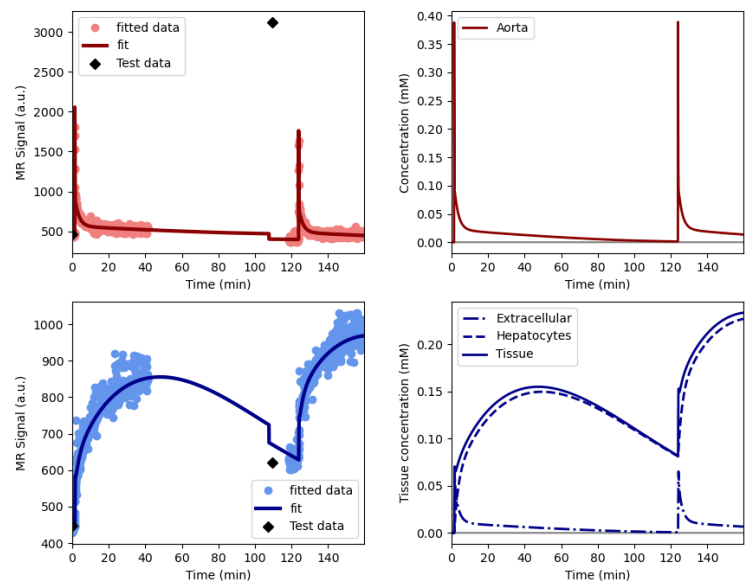


Figure 1.2: Signal-time curves for subject SHF-003 at the control visit.

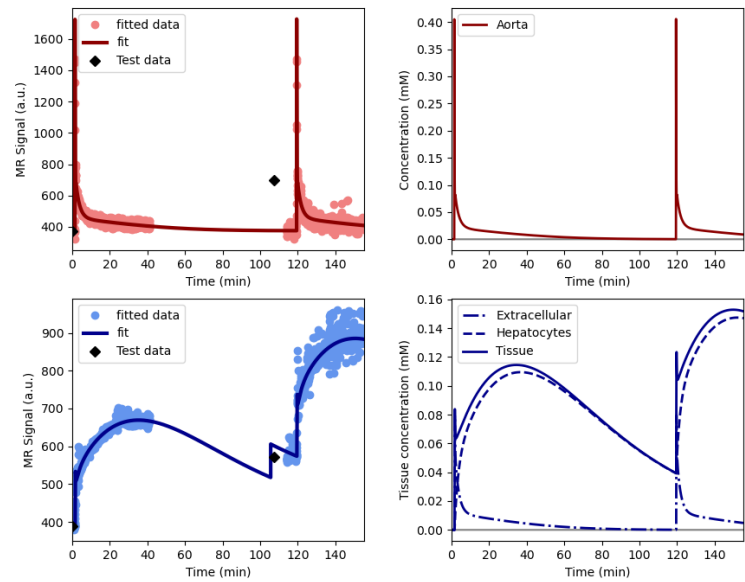


Figure 1.3: Signal-time curves for subject SHF-003 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	261.28	208.09	-20.36
AUC for CI (0-inf)	mM*sec	1200.89	732.74	-38.98
Biliary excretion rate	mL/min/100cm3	1.03	0.89	-13.16
Biliary tissue excretion rate	/min	0.02	0.02	-1.22
Final biliary excretion rate	mL/min/100cm3	0.92	0.82	-10.97
Final hepatocellular uptake rate	mL/min/100cm3	17.85	15.13	-15.23
Gut dispersion	%	74.3	60.07	-19.15
Gut mean transit time	sec	19.82	18.58	-6.29
Hepatocellular mean transit time	min	50.39	51.02	1.23
Hepatocellular tissue uptake rate	/min	0.62	0.48	-22.58
Hepatocellular uptake rate	mL/min/100cm3	16.35	14.3	-12.56
Initial biliary excretion rate	mL/min/100cm3	1.16	0.97	-15.75
Initial hepatocellular uptake rate	mL/min/100cm3	14.85	13.46	-9.35
Liver T1-MOLLI at 45min	sec	0.36	0.39	9.79
Liver T1-MOLLI at baseline	sec	0.72	0.76	4.8
Liver T1-MOLLI at scan 2	sec	0.47	0.59	26.68
Liver extracellular volume fraction	mL/100cm3	26.56	30.0	12.94
Liver plasma clearance	L/min	0.16	0.13	-20.41
RE for R1l at 20min	%	91.69	77.8	-15.15
RE for Sl at 20min	%	79.42	69.4	-12.62

Table 1.6: Values for liver of subject SHF-003

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	47.89	40.52	-15.38
AUC for Cb (0-inf)	mM*sec	92.03	59.94	-34.87
Body extraction fraction	%	4.39	5.79	31.83
Cardiac output	L/min	6.89	8.19	18.96
Heart-lung dispersion	%	47.15	40.31	-14.5
Heart-lung mean transit time	sec	22.49	20.57	-8.5
Organs blood mean transit time	sec	20.8	17.52	-15.76
Organs extraction fraction	%	19.73	17.69	-10.33
Organs extravascular mean transit time	min	7.69	6.72	-12.53
RE for R1b at 20min	%	18.34	14.91	-18.69
RE for Sb at 20min	%	9.31	11.43	22.77

Table 1.7: Values for aorta of subject SHF-003

1.4.2. Subject SHF-008

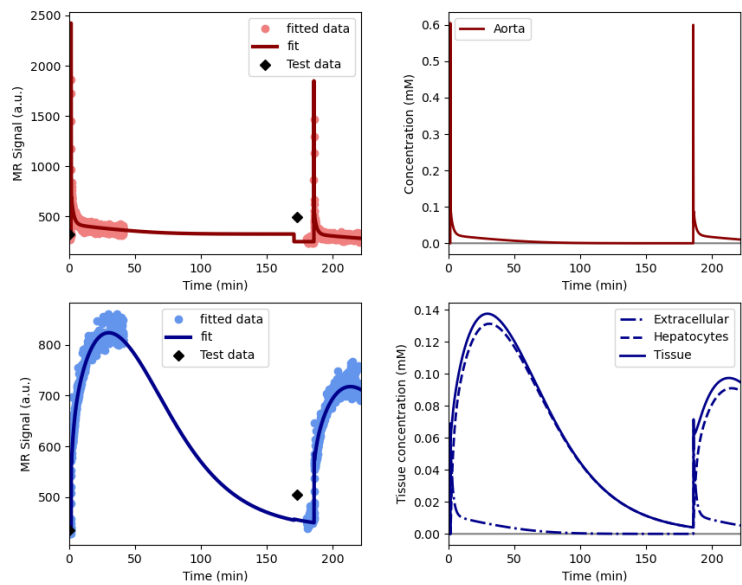


Figure 1.4: Signal-time curves for subject SHF-008 at the control visit.

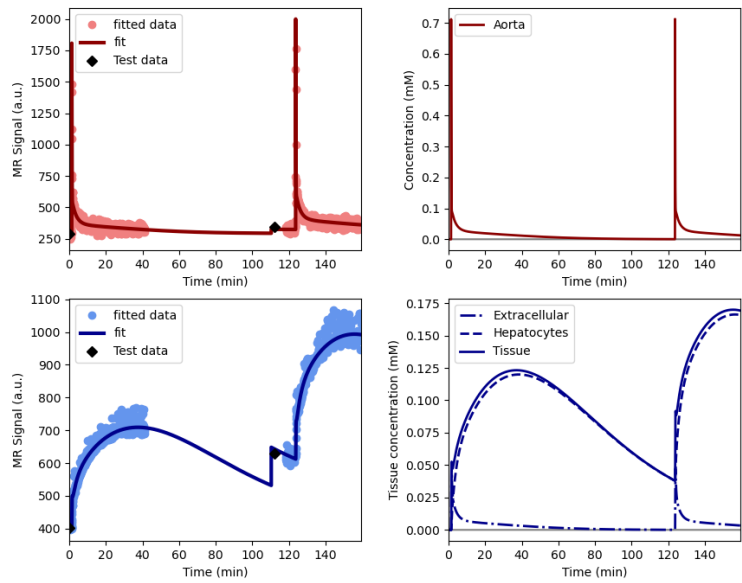


Figure 1.5: Signal-time curves for subject SHF-008 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	228.02	218.76	-4.06
AUC for CI (0-inf)	mM*sec	741.47	794.55	7.16
Biliary excretion rate	mL/min/100cm ³	1.43	1.6	12.12
Biliary tissue excretion rate	/min	0.03	0.02	-25.86
Final biliary excretion rate	mL/min/100cm ³	1.39	1.58	13.4
Final hepatocellular uptake rate	mL/min/100cm ³	11.64	14.5	24.52
Gut dispersion	%	70.5	81.26	15.26
Gut mean transit time	sec	21.83	21.16	-3.06
Hepatocellular mean transit time	min	33.94	45.78	34.88
Hepatocellular tissue uptake rate	/min	0.54	0.88	61.19
Hepatocellular uptake rate	mL/min/100cm ³	15.38	12.83	-16.59
Initial biliary excretion rate	mL/min/100cm ³	1.47	1.63	10.8
Initial hepatocellular uptake rate	mL/min/100cm ³	19.13	11.17	-41.6
Liver T1-MOLLI at 45min	sec	0.42	0.43	1.39
Liver T1-MOLLI at baseline	sec	0.79	0.74	-5.76
Liver T1-MOLLI at scan 2	sec	0.67	0.56	-16.31
Liver extracellular volume fraction	mL/100cm ³	28.32	14.66	-48.25
Liver plasma clearance	L/min	0.13	0.1	-22.77
RE for R1l at 20min	%	100.41	80.31	-20.01
RE for Sl at 20min	%	83.08	71.59	-13.83

Table 1.8: Values for liver of subject SHF-008

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	41.69	53.46	28.24
AUC for Cb (0-inf)	mM*sec	66.08	83.66	26.61
Body extraction fraction	%	4.4	6.16	40.16
Cardiac output	L/min	6.53	3.68	-43.56
Heart-lung dispersion	%	30.26	40.89	35.13
Heart-lung mean transit time	sec	14.9	17.51	17.5
Organs blood mean transit time	sec	16.33	32.22	97.31
Organs extraction fraction	%	17.02	19.57	14.98
Organs extravascular mean transit time	min	6.07	6.57	8.28
RE for R1b at 20min	%	20.62	18.23	-11.56
RE for Sb at 20min	%	32.13	17.41	-45.82

Table 1.9: Values for aorta of subject SHF-008

1.4.3. Subject SHF-011

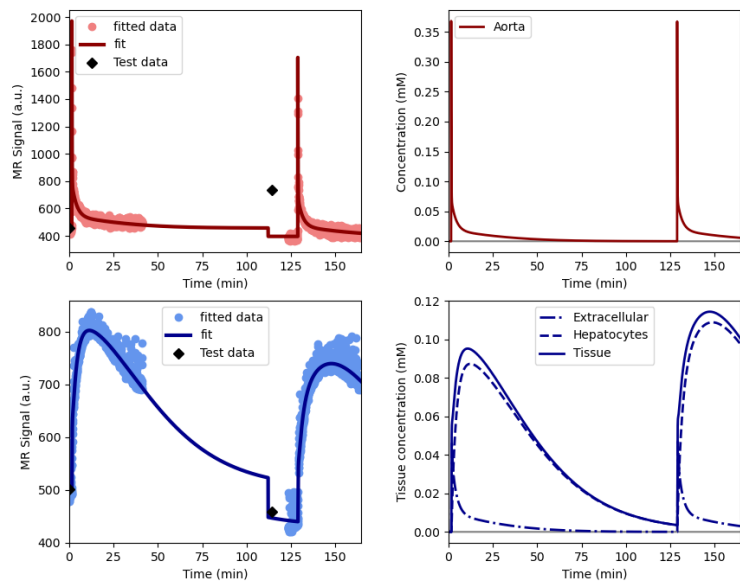


Figure 1.6: Signal-time curves for subject SHF-011 at the control visit.

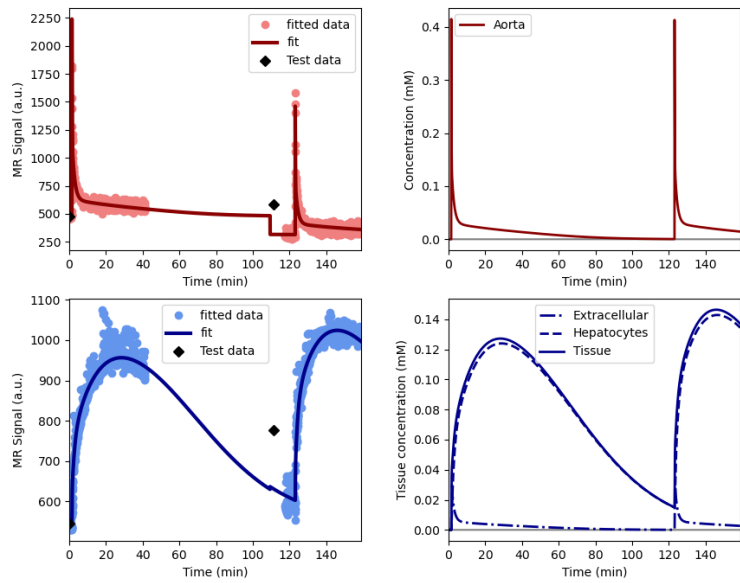


Figure 1.7: Signal-time curves for subject SHF-011 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	193.78	234.01	20.76
AUC for CI (0-inf)	mM*sec	327.52	649.14	98.2
Biliary excretion rate	mL/min/100cm3	2.31	3.52	52.77
Biliary tissue excretion rate	/min	0.05	0.04	-9.09
Final biliary excretion rate	mL/min/100cm3	1.67	3.8	126.76
Final hepatocellular uptake rate	mL/min/100cm3	20.62	19.29	-6.47
Gut dispersion	%	76.06	82.05	7.88
Gut mean transit time	sec	27.06	15.0	-44.56
Hepatocellular mean transit time	min	21.05	23.15	10.0
Hepatocellular tissue uptake rate	/min	0.77	1.72	122.63
Hepatocellular uptake rate	mL/min/100cm3	21.82	17.4	-20.25
Initial biliary excretion rate	mL/min/100cm3	3.71	3.29	-11.32
Initial hepatocellular uptake rate	mL/min/100cm3	23.03	15.52	-32.6
Liver T1-MOLLI at 45min	sec	0.5	0.42	-14.81
Liver T1-MOLLI at baseline	sec	0.76	0.72	-5.54
Liver T1-MOLLI at scan 2	sec	0.71	0.49	-30.79
Liver extracellular volume fraction	mL/100cm3	28.3	10.14	-64.18
Liver plasma clearance	L/min	0.18	0.14	-23.4
RE for R1l at 20min	%	62.95	86.46	37.34
RE for Sl at 20min	%	53.57	75.55	41.03

Table 1.10: Values for liver of subject SHF-011

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	33.19	54.86	65.29
AUC for Cb (0-inf)	mM*sec	44.79	90.87	102.87
Body extraction fraction	%	6.91	5.89	-14.76
Cardiac output	L/min	5.71	3.28	-42.54
Heart-lung dispersion	%	56.13	56.72	1.05
Heart-lung mean transit time	sec	15.14	23.99	58.45
Organs blood mean transit time	sec	29.4	16.02	-45.51
Organs extraction fraction	%	13.82	29.42	112.83
Organs extravascular mean transit time	min	7.95	4.63	-41.83
RE for R1b at 20min	%	11.28	21.81	93.41
RE for Sb at 20min	%	16.33	17.16	5.07

Table 1.11: Values for aorta of subject SHF-011

1.4.4. Subject SHF-014

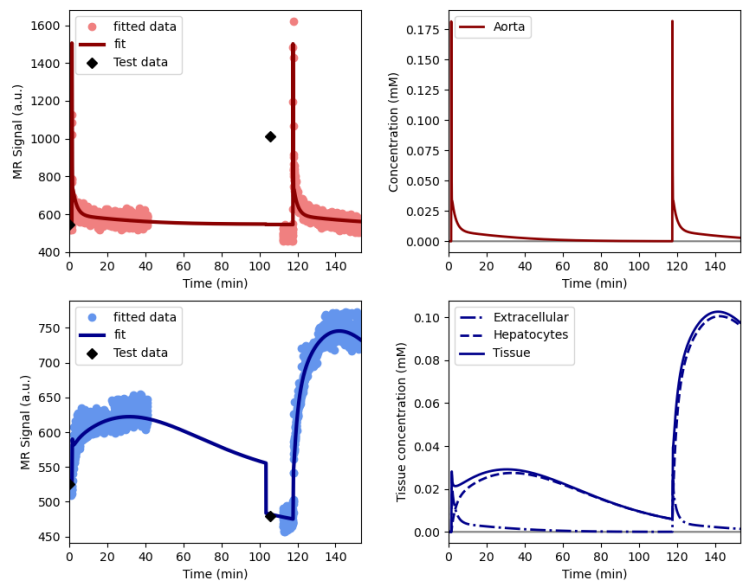


Figure 1.8: Signal-time curves for subject SHF-014 at the control visit.

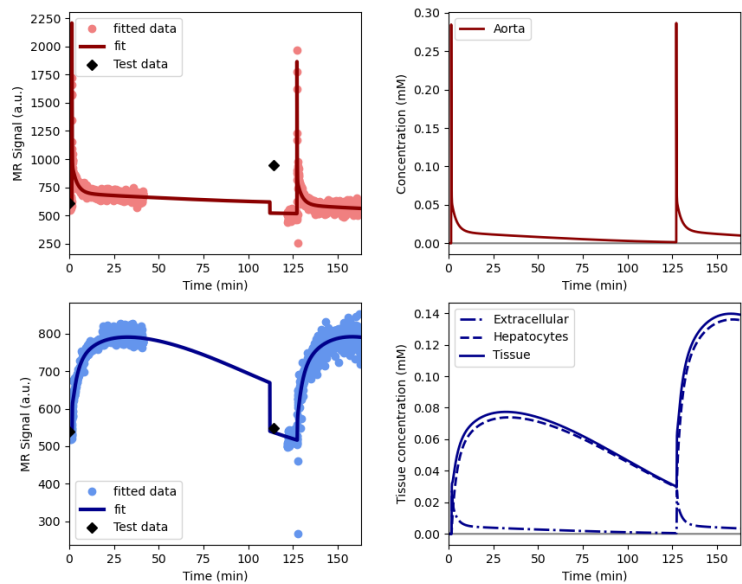


Figure 1.9: Signal-time curves for subject SHF-014 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	83.82	164.23	95.93
AUC for CI (0-inf)	mM*sec	156.81	554.88	253.85
Biliary excretion rate	mL/min/100cm ³	1.73	2.71	56.83
Biliary tissue excretion rate	/min	0.03	0.04	22.98
Final biliary excretion rate	mL/min/100cm ³	1.62	2.33	43.48
Final hepatocellular uptake rate	mL/min/100cm ³	43.19	24.92	-42.29
Gut dispersion	%	80.94	64.64	-20.14
Gut mean transit time	sec	20.54	27.97	36.15
Hepatocellular mean transit time	min	29.98	24.38	-18.69
Hepatocellular tissue uptake rate	/min	0.97	1.11	14.13
Hepatocellular uptake rate	mL/min/100cm ³	25.69	20.63	-19.69
Initial biliary excretion rate	mL/min/100cm ³	1.85	3.25	75.46
Initial hepatocellular uptake rate	mL/min/100cm ³	8.19	16.34	99.5
Liver T1-MOLLI at 45min	sec	0.56	0.72	29.04
Liver T1-MOLLI at baseline	sec	0.73	0.72	-1.47
Liver T1-MOLLI at scan 2	sec	0.69	0.56	-19.85
Liver extracellular volume fraction	mL/100cm ³	26.49	18.64	-29.63
Liver plasma clearance	L/min	0.29	0.24	-15.47
RE for R1l at 20min	%	18.98	51.87	173.32
RE for Sl at 20min	%	14.92	53.22	256.76

Table 1.12: Values for liver of subject SHF-014

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	15.95	33.14	107.75
AUC for Cb (0-inf)	mM*sec	21.92	69.62	217.55
Body extraction fraction	%	8.14	3.52	-56.75
Cardiac output	L/min	15.25	10.54	-30.85
Heart-lung dispersion	%	42.36	51.9	22.54
Heart-lung mean transit time	sec	22.85	17.03	-25.49
Organs blood mean transit time	sec	23.73	29.17	22.9
Organs extraction fraction	%	19.12	16.37	-14.38
Organs extravascular mean transit time	min	7.13	8.72	22.36
RE for R1b at 20min	%	5.5	12.79	132.51
RE for Sb at 20min	%	5.76	8.69	50.71

Table 1.13: Values for aorta of subject SHF-014

1.4.5. Subject SHF-021

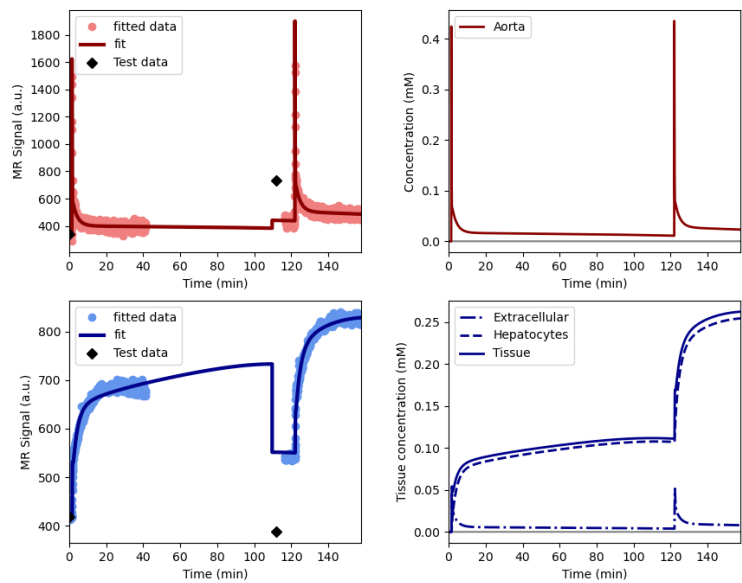


Figure 1.10: Signal-time curves for subject SHF-021 at the control visit.

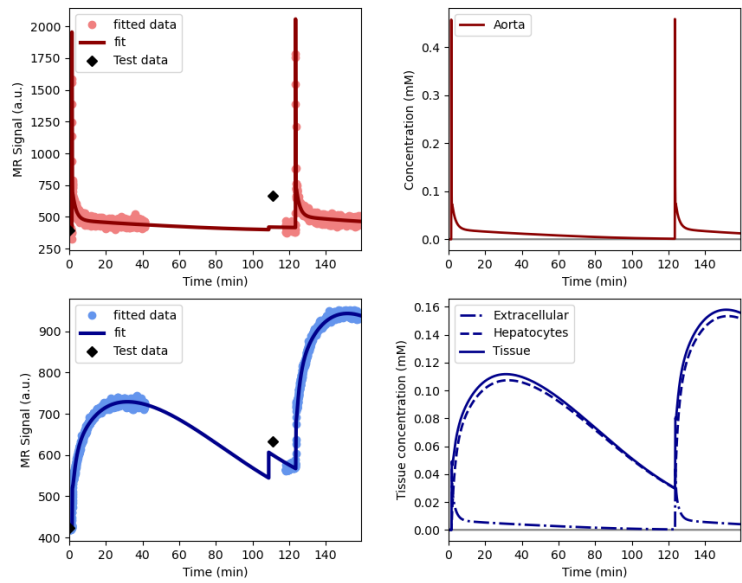


Figure 1.11: Signal-time curves for subject SHF-021 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	206.98	217.23	4.95
AUC for CI (0-inf)	mM*sec	1103.91	696.39	-36.92
Biliary excretion rate	mL/min/100cm3	3.56	2.65	-25.65
Biliary tissue excretion rate	/min	0.05	0.04	-27.22
Final biliary excretion rate	mL/min/100cm3	3.16	2.45	-22.35
Final hepatocellular uptake rate	mL/min/100cm3	29.67	22.68	-23.56
Gut dispersion	%	61.96	66.41	7.17
Gut mean transit time	sec	21.91	21.76	-0.68
Hepatocellular mean transit time	min	18.41	25.29	37.39
Hepatocellular tissue uptake rate	/min	1.24	1.11	-10.66
Hepatocellular uptake rate	mL/min/100cm3	23.53	20.16	-14.33
Initial biliary excretion rate	mL/min/100cm3	4.09	2.88	-29.52
Initial hepatocellular uptake rate	mL/min/100cm3	17.4	17.64	1.4
Liver T1-MOLLI at 45min	sec	0.47	0.43	-8.89
Liver T1-MOLLI at baseline	sec	0.8	0.78	-2.93
Liver T1-MOLLI at scan 2	sec	0.64	0.56	-12.25
Liver extracellular volume fraction	mL/100cm3	18.92	18.14	-4.11
Liver plasma clearance	L/min	0.23	0.18	-21.48
RE for R1l at 20min	%	68.68	80.76	17.58
RE for Sl at 20min	%	63.99	69.62	8.79

Table 1.14: Values for liver of subject SHF-021

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	43.4	42.48	-2.13
AUC for Cb (0-inf)	mM*sec	153.66	81.1	-47.22
Body extraction fraction	%	1.02	4.19	310.22
Cardiac output	L/min	6.05	6.66	10.05
Heart-lung dispersion	%	41.62	39.93	-4.08
Heart-lung mean transit time	sec	21.47	18.46	-14.03
Organs blood mean transit time	sec	29.53	22.63	-23.37
Organs extraction fraction	%	22.57	20.86	-7.56
Organs extravascular mean transit time	min	10.31	6.51	-36.88
RE for R1b at 20min	%	17.16	16.76	-2.34
RE for Sb at 20min	%	17.32	13.91	-19.7

Table 1.15: Values for aorta of subject SHF-021

1.4.6. Subject SHF-022

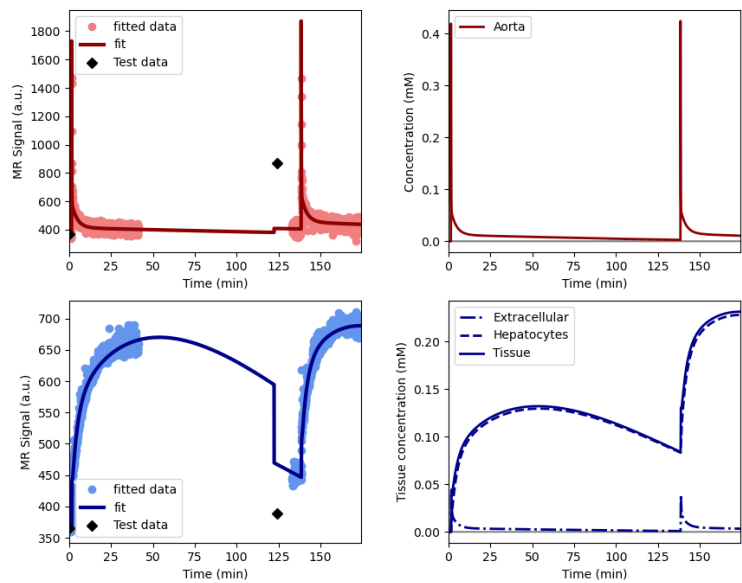


Figure 1.12: Signal-time curves for subject SHF-022 at the control visit.

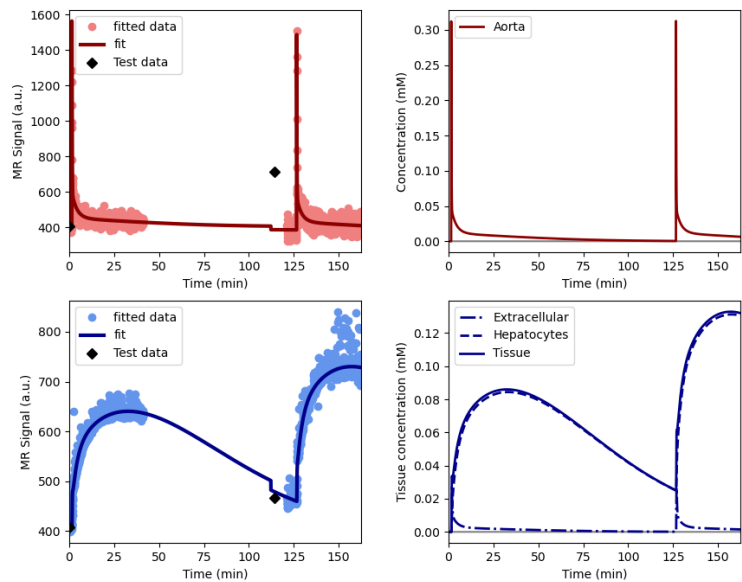


Figure 1.13: Signal-time curves for subject SHF-022 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	235.42	168.98	-28.22
AUC for Cl (0-inf)	mM*sec	1168.54	554.69	-52.53
Biliary excretion rate	mL/min/100cm3	1.81	2.28	26.37
Biliary tissue excretion rate	/min	0.03	0.03	12.34
Final biliary excretion rate	mL/min/100cm3	1.73	2.04	17.87
Final hepatocellular uptake rate	mL/min/100cm3	29.93	27.45	-8.26
Gut dispersion	%	69.61	62.2	-10.64
Gut mean transit time	sec	17.65	19.54	10.74
Hepatocellular mean transit time	min	37.93	33.77	-10.98
Hepatocellular tissue uptake rate	/min	1.47	1.88	27.92
Hepatocellular uptake rate	mL/min/100cm3	25.42	23.69	-6.81
Initial biliary excretion rate	mL/min/100cm3	1.88	2.58	37.11
Initial hepatocellular uptake rate	mL/min/100cm3	20.92	19.93	-4.72
Liver T1-MOLLI at 45min	sec	0.37	0.43	16.95
Liver T1-MOLLI at baseline	sec	0.76	0.78	3.03
Liver T1-MOLLI at scan 2	sec	0.55	0.65	17.9
Liver extracellular volume fraction	mL/100cm3	17.33	12.63	-27.15
Liver plasma clearance	L/min	0.29	0.28	-3.99
RE for R1l at 20min	%	85.42	62.69	-26.61
RE for Sl at 20min	%	75.22	54.98	-26.9

Table 1.16: Values for liver of subject SHF-022

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	32.35	25.71	-20.54
AUC for Cb (0-inf)	mM*sec	79.06	44.35	-43.9
Body extraction fraction	%	2.85	7.15	150.66
Cardiac output	L/min	8.51	7.21	-15.26
Heart-lung dispersion	%	46.09	45.72	-0.81
Heart-lung mean transit time	sec	13.34	22.56	69.18
Organs blood mean transit time	sec	32.42	44.51	37.31
Organs extraction fraction	%	15.6	22.52	44.39
Organs extravascular mean transit time	min	13.3	8.29	-37.63
RE for R1b at 20min	%	11.49	9.2	-19.96
RE for Sb at 20min	%	17.51	10.79	-38.34

Table 1.17: Values for aorta of subject SHF-022

2

Secondary results

2.1. Diurnal variation

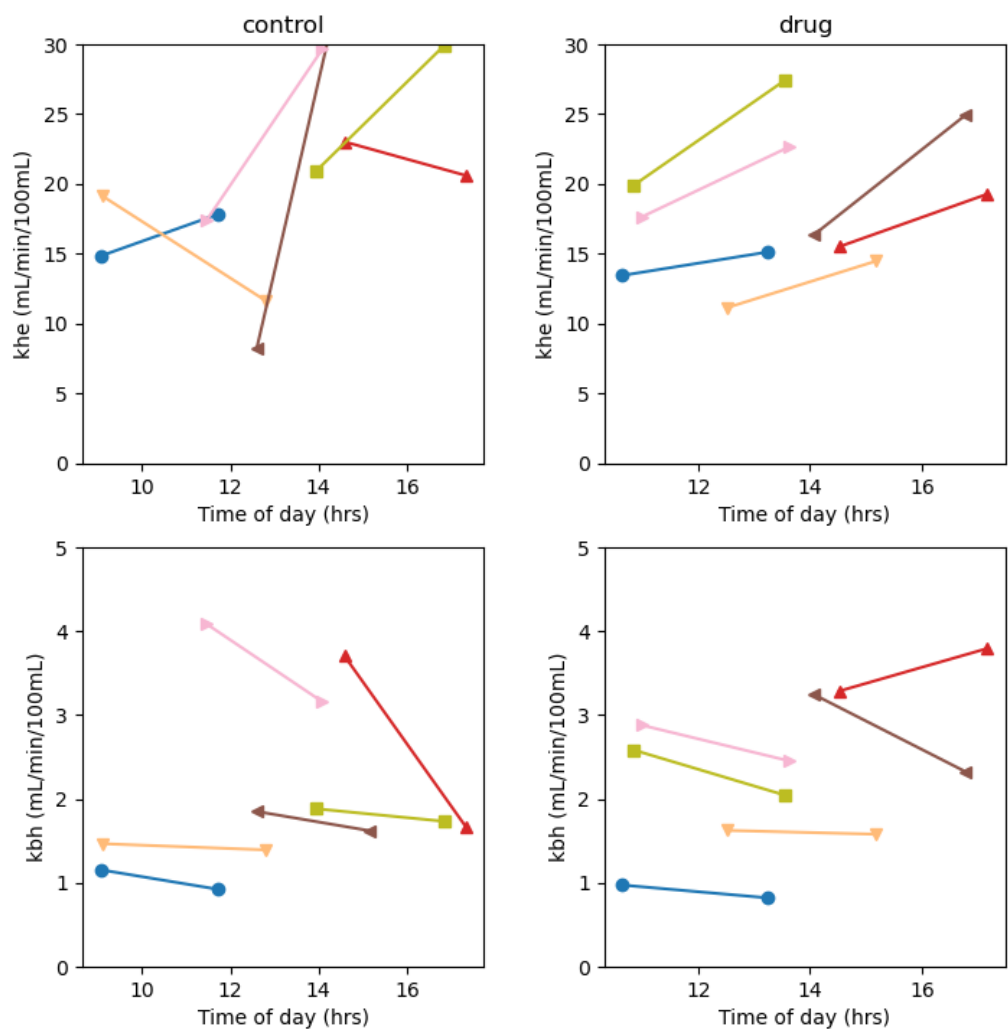


Figure 2.1: Intra-day changes in hepatocellular uptake (k_{he} , top row) and biliary excretion (k_{bh} , bottom row) of gadoxetate at for the control (left column) and treatment (right column). Full lines connect values taken in the same subject at the same day.